An upstream placebo shows that reservoir siting explains the apparent drought buffering of seasonal reservoirs in Chile's megadrought

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Key Points:

- Apparent drought buffering is as strong above dams as below, reflecting where seasonal dams sit, not what they do
- The powered null excludes large operational buffering; only carryover-capable reservoirs show an exploratory downstream-specific signal
- With storage stepping down as supply falls, expanding seasonal storage is supply-blind and adaptation must shift toward managing demand

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Abstract

Much of what is attributed to reservoir drought buffering may be a measurable siting confound. We treat reservoir presence as a basin-level intervention in Chile's megadrought and separate siting from operation with an upstream/downstream placebo. Across streamflow, the apparent buffering is as strong upstream as downstream (slopes −0.20 versus −0.16; pooled gap $p = 0.39$, paired contrast $p = 0.72$), and it collapses once climate regions carry their own baseline transmission (residual −0.05, $p = 0.65$). Over the first decade of the megadrought the null is powered against large effects, excluding operational buffering above 46 to 59% of baseline transmission. But the fleet is storage-limited: in an exploratory split, the seven carryover-capable reservoirs (capacity at least half a year of downstream flow) show downstream-specific buffering (contrast −0.23, $p = 0.014$) that the seasonal majority does not; its direction is robust to the capacity classification and to single-basin exclusion, its significance to neither, so carryover reservoirs are candidates for genuine buffering rather than a confirmed exception. Dammed basins hold 4.5 times more cropland than matched controls, a siting legacy with no detected divergence within power; water-rights accrual is consistent with no effect but underpowered, a genuine null on new state grants that cannot see market reallocation in administratively closed basins. The storage band steps down at 2010 and the carryover floor keeps eroding. The upstream placebo is a portable method. For the storage-limited fleet, within these bounds, adaptation must shift toward managing demand.

Plain Language Summary

When drought strikes, governments often respond by building more reservoirs, on the assumption that storing water cushions communities when rainfall fails. But dams are built where water is already scarce, so basins with dams differ from basins without them even before any drought. We separated the effect of the dam from the effect of its location by comparing river gauges above and below each dam, and by matching dry dammed basins to similar undammed ones across Chile's decade long megadrought. Once location is accounted for, the smaller, seasonal reservoirs that dominate the fleet show no measurable buffer against multi year drought, and the apparent buffering reflects where dams are built, not what they do. But the large carryover reservoirs, the few that hold at least half a year of flow, do show a downstream buffering signal, so the capacity to store water across years matters. With the water available to store falling, our results suggest that building more small reservoirs is not the answer; managing how water is used is.

1 Introduction

A persistent obstacle separates what we think reservoirs do during drought from what they actually do: dams are built where water is already scarce, so dammed basins differ systematically from undammed ones before any drought strikes. The short-term hydrological benefit of storage is documented (Wu et al., 2017, 2018), and model-based counterfactuals, in which a reservoir's effect is read against reconstructed or simulated unregulated flow, find that regulation attenuates hydrological drought downstream (Wan et al., 2017; Wu et al., 2019, 2021). Whether this buffering extends to multi-year drought, however, remains contested, because those counterfactuals are themselves model-dependent: they can embed assumptions about where the dam sits and about the unregulated baseline it is measured against. The distinction matters. If buffering is real, expanding storage is a rational adaptation to intensifying drought. If it is siting, then the reflexive global response, more dams, invests in the wrong mechanism.

The reservoir-effect hypothesis (Di Baldassarre et al., 2018) formalizes the concern: greater supply invites expansion, so demand grows to meet and then exceed each increment of storage (the supply-demand cycle), and prolonged reliance on stored abundance erodes the incentive to adapt, magnifying the damage when drought finally outlasts the reservoir. Consistent with this, global demand has outpaced storage for decades and most reservoirs now fail to refill (Li et al., 2023; Wang et al., 2024), yet whether these feedbacks dominate, or are outweighed by the protection dams provide, remains debated (Kellner, 2021). What the empirical literature still lacks is a counterfactual that separates a reservoir effect from the siting decision without leaning on a modeled unregulated baseline. Paired-gauge comparisons, which read a dam's effect against an unregulated reference reach, have been applied at national scale to flow-regime alteration (Chalise et al., 2021, 2023), but not to the transmission of multi-year meteorological drought conditional on the deficit actually experienced, the quantity that determines whether reservoirs buffer the droughts that matter.

We address this with a within-basin falsification test. If a dam buffers drought, streamflow transmission from meteorological to hydrological drought should flatten only below the dam, where regulation operates. If it flattens equally at the unregulated upstream reaches of the same basins, the attenuation is siting, not regulation. This upstream/downstream placebo is the paper's central identification strategy. We complement it with a matched cross-basin comparison of each dammed basin against hydroclimatically comparable undammed ones, evaluating both against the drought each basin actually experienced rather than calendar time, breaking the collinearity between the megadrought onset and the treatment clock.

Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an uninterrupted run of dry years shading into structural aridification with a documented anthropogenic component (Boisier et al., 2018; R. Garreaud et al., 2025; R. D. Garreaud et al., 2017; Vicente-Serrano et al., 2019; Zambrano et al., 2025). The 1981 Water Code makes water-use rights perpetual private property that the state cannot readily curtail, so the policy reflex has been supply-side, new dams and inter-basin transfers, alongside market incentives that concentrated water in high-value export crops (Barría et al., 2021; Reade Malagueño & D'Odorico, 2024; Rivera et al., 2016). In the snow-fed basins of central Chile demand already outruns supply in roughly three years of every ten (Webb et al., 2021). Chile thus exhibits every ingredient of the reservoir effect, making it a most-likely case for detecting one if it operates, and its short, steep Andean catchments, with streamflow gauges above and below major dams, provide the physical setup the upstream placebo requires, a configuration present across the world's regulated mountain rivers from the Sierra Nevada to the European Alps to the African Rift.

We apply this design across the streamflow, cropland, and water-rights pathways, together with an evapotranspiration pathway that we evaluate against the known limitations of the available product. Throughout, outcomes are evaluated at annual to multi-year accumulations, the timescale of structural drought, not of seasonal operation.

Our objective is to adjudicate two competing accounts of the attenuated drought transmission that dammed basins have been reported to show, and to bound what the data can and cannot decide between them. Under the operational-buffering account, that attenuation is a regulation effect: transmission should flatten specifically below dams and not at the unregulated reaches above them, should persist when dammed basins are compared within their own climate regime, and should concentrate in the reservoirs whose storage is large enough to carry water across years. Under the siting account, the same attenuation is a property of where dams are built: it should be equally present upstream, and should dissolve once the aridity gradient along which reservoirs are sited is removed. The reservoir-effect hypothesis adds a demand-side prediction that runs against buffering altogether: if storage induces the expansion it is built to serve, dammed basins should accumulate irrigated land and water claims faster through a prolonged drought, steepening their deficit-to-impact response rather than flattening it. We test each prediction with pre-specified falsifiers, and, because a failure to reject is only as informative as the effect size the design can exclude, we report the minimum detectable effect alongside every estimate.

2 Materials and Methods

2.1 Study area, units, and treatment

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates (Vásquez et al., 2025), over 2005-2024, when reservoir storage, meteorological forcing, and satellite outcomes jointly exist. The design is a matched comparison of dammed versus undammed sub-watersheds. Because most treated reservoirs (18 of 24) predate the storage record, treatment is effectively time-invariant and a staggered-adoption event study is infeasible; identification rests on a matched cross-section conditioned on meteorological forcing rather than on calendar time. The estimand is a reservoir's operational effect *conditional on the covariates that govern where reservoirs are sited*, isolating *operation* from *siting* (Di Baldassarre et al., 2018).

The analysis grain is the DGA sub-watershed (467 units), chosen over the watershed grain because reservoirs sit in Chile's largest main-stem basins and the finer grain supplies more controls inside the treated size range. A sub-watershed is treated if it contains 1 of the 26 monitored reservoirs, giving 24 treated units, of which 21 survive elevation common support. The reservoirs are predominantly irrigation (20 of 26), and the streamflow result is unchanged restricted to irrigation-only basins (Supplementary Table S11). Untreated sub-watersheds inside a reservoir's watershed are removed as contaminated (45 units), leaving 398 clean controls, which defuses the most direct downstream-spillover (SUTVA) violation; no matched pair shares a watershed (Supplementary Table S15). Because "undammed" is defined against the 26 monitored reservoirs, the matched controls are additionally screened against the full 1,370-impoundment national inventory and the results re-fit with the entropy balance re-estimated on the dam-free control pool (Supplementary Table S53).

2.2 Data

Reservoir storage. Monthly storage for 26 monitored reservoirs (DGA, 2005–2026; we analyse complete calendar years 2005–2024) with per-reservoir capacity in hm^3; because storage and capacity share units, percent-of-capacity is the cross-reservoir-comparable state variable. The band trends are re-fit three further ways, since capacities span three orders of magnitude and several reservoirs enter the panel after 2005 with filling phases that static fixed effects do not absorb: weighted by capacity and as fleet-total volumes (Supplementary Table S44), on the coverage-stable subset alone (Supplementary Table S51), and, because a descriptive trend offers no treatment assignment to permute, with every coefficient confirmed by a restricted wild cluster bootstrap using reservoir-level Rademacher weights (Supplementary Table S54).
For the capacity heterogeneity test the buffering sample (17 treated basins with a usable downstream gauge) is split at a carryover ratio of 0.5, capacity divided by mean annual downstream flow in years. Because that denominator is regulated and consumed, the ratio is recomputed with unregulated upstream-gauge flow (Supplementary Table S40); and because the class mixes irrigation reservoirs with two dams operated also for hydropower (Colbún, Lago Laja), and seven units are few enough that one basin can move a permutation p across conventional thresholds, the contrast is re-estimated excluding that pair, under leave-one-out, and on strictly paired basins

(Supplementary Tables S39, S42). That split is exploratory and reviewer-prompted, carrying no multiplicity adjustment.

Meteorological forcing. SPEI at 12-month accumulation, from CHIRPS precipitation and CHIRTS temperature standardized on 1991–2020 (Sergio M. Vicente-Serrano et al., 2010), is the exogenous dose; CHIRPS-derived indices are validated for Chilean drought monitoring (Meza, 2013; Oertel et al., 2020; Zambrano et al., 2017). The 12-month accumulation matches the annual, snowmelt-fed cycle of central Chile and the inter-seasonal carryover irrigation reservoirs bridge; estimates are insensitive to 6- and 24-month alternatives, and the window does not resolve 1 to 3 month buffering, a boundary of our conclusions (Wu et al., 2017, 2018). Conditioning on forcing breaks the collinearity between megadrought onset and treatment clock.

Hydrological drought. Daily streamflow from the DGA network (2000–2020), compiled by CR2 (https://cr2.cl), aggregated to SSI-12 (Lema et al., 2025; Sergio M. Vicente-Serrano et al., 2012), the regulated-flow analogue of SPEI-12 and the placebo outcome; gauges are classified upstream or downstream of each dam by SRTM elevation. Missing values are never imputed: monthly means require 20 valid days, the 12-month accumulation propagates any missing month, gauges need 96 monthly SSI values, and stations 10 years of record extending past 2010. Gaps therefore shrink the sample rather than distort the index, and their drought-conditional structure is quantified overall and by gauge class (Supplementary Table S50). SSI is standardized on the common 2000–2020 window so all gauges share one reference; because that window is megadrought-dominated and shorter than the 30-year guideline, estimates are re-derived on the 1990–2020 baseline, which the 1900-onset record permits (Supplementary Table S49). Because SSI normalizes each gauge, the slope could divide out dam-induced variance compression, so models are re-estimated on log 12-month mean flow and the compression premise tested directly (Supplementary Table S19).

Land use, water rights, and auxiliary data. Annual cropland area comes from MapBiomas Chile Collection 2 (30 m, 1999–2024) and evapotranspiration from MODIS MOD16 (8-day, ~500 m, 2001–2024), whole-basin and for an orchard stratum from the CIREN/ODEPA cadastre; both track drought-driven agricultural losses in Chile (Zambrano et al., 2018). Consumptive rights from the DGA registry, which records each right's grant date, flow, and capture point, are assigned to sub-watersheds as an annual cumulative stock; because the registry carries severe volume errors, the primary measure is rights density (count per 100 km^2), with a winsorized volume series insensitive to threshold (Supplementary Table S8) and rank-based tests on raw volumes (Supplementary Table S18; both in Supplementary Fig. S6). ERA5-Land snow water equivalent (0.1°, 2000–2026) tests the snowmelt confound on the placebo estimand through a forcing-by-snowpack adjustment, a low-snow subsample, a snow-year interaction, and an early/late split (Supplementary Table S12); its bias is multiplicative, so standardization removes it, unlike the land-cover-dependent MOD16 ET bias that disqualifies the ET outcomes (Supplementary Table S21). DGA well hydrographs (2000–2021, 213 wells) give per-basin depletion rates bounding the groundwater-substitution confound (Supplementary Table S10).

Matching covariates. Baseline aridity (P/PET over the 1991–2020 WMO normal), watershed area, mean elevation (SRTM), and Köppen–Geiger class, by zonal statistics; recomputing aridity on the strictly pre-drought 1991–2009 window is essentially identical ($r = 0.998$), so the matching does not lean on drought-period variation (Supplementary Table S13).

2.3 Matched control construction

We estimate the average treatment effect on the treated (ATT) by *entropy balancing*, exact-matched within Köppen main group and balancing log(area) and mean elevation; latitude is left untargeted because it fights climate class across the Andes–Patagonia span, and balancing elevation raised the effective control sample from 9 to 91 while balancing latitude as a by-product. The procedure retains 21 of 24 treated units with essentially perfect balance (standardized mean differences falling from 1.18/0.37 to 0.000) against 244 in-window controls. Baseline aridity is a genuine confound (median P/PET 0.26 treated versus 0.80 control) but is *not* a balance target, because hard-balancing collapses the effective control sample from 91 to 19; the residual imbalance (standardized mean difference 0.17) is absorbed by the doubly-robust outcome model, an interpolation within support since all treated basins lie inside the control aridity range (Supplementary Fig. S8), and weighting-only and overlap-subset fits leave the central nulls unchanged (Supplementary Tables S6, S7). Baseline cropland is deliberately not a balance target either, since cropland concentration is plausibly a consequence of siting and sits adjacent to the demand outcomes, so balancing it would condition on a mediator; its functional form is instead tested by metric invariance on log cropland share (Supplementary Table S20). Because transmission can be non-linear in aridity, the model is also re-fit with quadratic aridity-by-forcing interactions (Supplementary Table S52). The strict hard-balanced fit (log-aridity standardized mean difference 0.000, effective sample 19) is a primary sensitivity, reported with weight-concentration diagnostics and leave-one-out re-fits (Supplementary Tables S17, S45); at that sample it corroborates rather than powers the result. The decisive evidence, the within-basin placebo, the within-region comparison, and randomization inference, uses no outcome-model aridity adjustment, and coarsened exact matching and nearest-neighbour matching on the identical sample give the same effects.

2.4 Estimands, estimators, and inference

Matched ATT and transmission slope. Each outcome is estimated weighting only, regression only, and doubly-robust (entropy weights plus an outcome model in log(area), elevation, and log(aridity)), so functional-form sensitivity is visible; covariates are centred at the treated mean and standard errors are HC3-robust. To remove megadrought *exposure*, concentrated in exactly the arid basins where reservoirs sit, the outcome for that machinery is the per-basin *transmission slope*, the regression of an annual outcome on annual SPEI-12; the vulnerability hypothesis predicts a *steeper* slope in dammed basins. Table 1 carries two level-change ATTs, cropland-area expansion (ha km^{-2} added) and water-rights accrual. A third outcome, reservoir ET buffering, applies the slope machinery to orchard-cell MOD16 ET and carries no evidential weight, since MOD16 underestimates ET over irrigated land and biases that gap toward zero.

Forcing-interacted difference-in-differences. On the observed area and ET panels we fit

$$y_{it} = \beta\,(\text{dammed}_i \times \text{SPEI}_{it}) + \gamma\,\text{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with sub-watershed (α_i) and year (τ_t) fixed effects and entropy weights. Because treatment is time-invariant the dose is meteorological SPEI rather than calendar year: year effects absorb the common megadrought shock and β is the differential deficit-to-impact slope, which the siting confound biases far less than it biases levels; the vulnerability hypothesis predicts $\beta > 0$. For slow-moving structural outcomes such as perennial cropland this tests only the annual drought-response differential, and cumulative expansion is instead tested by a dynamic event study (dammed $\times$ year, reference 2009) with bounded pre- and post-2010 trend tests. Because year effects absorb the common temporal component, β is identified from within-year spatial variation in SPEI; as a sensitivity retaining the temporal signal, the streamflow models are re-fit with year effects replaced by a cubic national time trend (Supplementary Table S47).

Bounding the nulls. Because the headline results are nulls, we bound the effect rather than rely on failure to reject: a two one-sided test declaring equivalence when the 90% interval lies within ±25% of baseline transmission, plus the threshold-free minimum detectable effect (MDE; 80% power, $\alpha = 0.05$). Since no hydrological standard defines a negligible reservoir effect, the margin is anchored in operational drought classification, whose severity categories are 0.5 standardized units wide (McKee et al., 1993; Sergio M. Vicente-Serrano et al., 2010), so a 25% slope change perturbs the propagated index by less than one category, though it still translates to a materially non-trivial volume (Supplementary Table S16). MDEs and equivalence intervals use the permutation-null standard deviation, since cluster-robust standard errors over-reject at ~21 clusters. A positive control confirms the null is informative: injected buffering is recovered one-to-one, with detection once it exceeds the MDE.

Inference. With ~21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is *randomization inference*: treatment is permuted among units within Köppen $\times$ aridity-tercile strata (999 permutations). Residual spatial dependence is assessed with Moran's I (Supplementary Table S9); where present, notably the water-rights outcomes, those outcomes are re-run with a permutation that swaps treatment at the watershed-block level, and this spatially valid inference governs where the two disagree. Residual confounding is probed with an aridity placebo relabeling the most-arid control tercile as pseudo-treated.

Siting-confound decomposition. A ladder on the same panel locates where any apparent buffering arises: (1) a naive pooled slope gap, what a conventional dammed-versus-undammed study reports; (2) unit and time fixed effects, unweighted; (3) the design estimator; (4) the design estimator with baseline transmission free to vary by Köppen region, reported with its own permutation null (Supplementary Table S26). The rungs are distinct controls, since the entropy balancing neither balances the residual within-region aridity gradient nor imposes a region-specific dose-response, so an apparent effect can survive rungs (1) to (3) and die at rung (4). The collapse from (3) to (4) is the between-region aridity confound; the residual at (4) is the regulation estimate.

2.5 Within-basin upstream/downstream placebo

The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated downstream reaches against unregulated upstream reaches of the *same* basins, which share climate, geology, and selection history: a genuine regulation effect implies a downstream-specific dose-response the upstream placebo does not reproduce. Of the 21 treated matched basins, 17 have a usable downstream gauge and enter the buffering sample; 15 also have an upstream gauge and form the paired contrast. The contrast $D = \beta_{\text{down}} - \beta_{\text{up}}$ is tested against its permutation distribution (Supplementary Table S25). Because control basins have no dam and therefore no natural gauge partition, the scheme holds panel composition fixed: each panel contains the treated basins' gauges of that class plus every control basin's full gauge set; in each draw treatment is reassigned at basin level within Köppen $\times$ aridity-tercile strata and the *same* draw applied to both panels before D is recomputed. Under the sharp null gauge class carries no dam-specific information, so D is centred on zero, and one draw across both panels preserves the between-panel dependence. Because a pseudo-treated control contributes an identical gauge set to both panels, the up/down heterogeneity real treated basins carry is absent from its null contribution, which could compress the null variance and make the test anti-conservative; that bias can only overstate significance, never manufacture a null, and both contrasts are re-tested under a variant null partitioning each pseudo-treated basin's gauges at its median elevation (Supplementary Table S48). Subgroup and covariate-adjusted contrasts inherit the same construction, and the contrast is also split by Köppen region.

The placebo's comparability assumptions are measured rather than assumed. Upstream reaches sit higher and drain smaller areas, so we quantify the elevation and area gradients among undammed control gauges and re-fit the placebo net of a SPEI-by-elevation interaction and, using each gauge's mean annual discharge as the cumulative-scale measure the local polygon area misses, net of a SPEI-by-log-discharge interaction (Supplementary Tables S2, S30, S31, S33); the linearity of the elevation sensitivity is checked across the ~2000 m snowline (Supplementary Table S20) and the bedrock-to-alluvial transition with 2.5 km local relief (Supplementary Table S22). Because gauge class is assigned from SRTM elevation, misclassification is bounded three ways: excluding downstream gauges whose DGA sub-cuenca is named for a different river and those within 100 m of the dam (Supplementary Table S32); validating against routed HydroRIVERS v1.0 topology, snapping every gauge and dam to its nearest reach, tracing the NEXT_DOWN path, and re-estimating on network-connected gauges only (Supplementary Table S46); and screening upstream gauges for cascade contamination against the national inventory at three tiers, below a *material* dam (DGA-monitored, registry-large, or wall height 15 m) inside their own sub-watershed, below the documented historical cascades, and below any material dam anywhere in their cuenca, the last over-flagging parallel tributaries to bound the worst case (Supplementary Table S43).

Four further specifications bound the remaining threats: a quadratic SPEI interaction and drought-month refits, testing whether the linear form masks an effect confined to extreme deficits (Supplementary Tables S28, S29); an early/late megadrought split, tested as a treat × SPEI × phase interaction under permutation (Supplementary Tables S24, S27); a down/up extraction gradient from MapBiomas cropland in gauge buffers, with the placebo re-fit net of a SPEI-by-local-cropland interaction carrying a treated-specific term, since drought-intensified downstream extraction could steepen the downstream slope and mask buffering (Supplementary Tables S36, S37); and a bound on the exposure misclassification implied by both gauge classes sharing the subcuenca-average SPEI (Supplementary Table S38). The paired contrast's own MDE is reported alongside (Supplementary Table S34).

3 Results

3.1 Reservoirs sit where water is already scarce

Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster in the arid centre and north, so a raw comparison would confuse the reservoir with its location. Two comparisons separate them (Figure 1). First, each dammed basin is compared only with undammed basins sharing its climate, size and elevation (21 of 24 dammed basins against an effective 91 of 244 controls; Supplementary Figs. S4, S5), both evaluated against the drought each experienced, so the contrast is impact per unit of deficit. Second, within each dammed basin, regulated reaches below the dam are compared against unregulated reaches above it. A genuine reservoir effect must appear in these comparisons. Across the fleet none does; the exception, developed below, is the carryover-capable minority. The controls are genuinely undammed: only 10 of the 244 hold a material dam from the 1,370-impoundment national inventory, and a re-balanced dam-free control pool leaves every streamflow estimate unchanged (Supplementary Table S53).

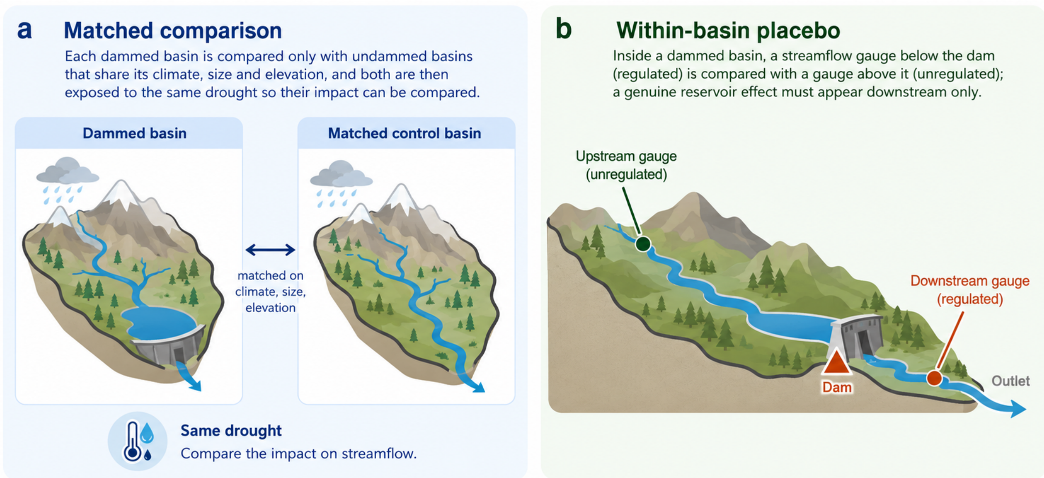

Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin against undammed basins sharing its climate, size and elevation, both exposed to the same drought. (b) Within-basin placebo: a gauge below the dam against one above it; a genuine reservoir effect must appear downstream only. Study-area map and matching diagnostics: Supplementary Figs. S4, S5.

3.2 Fleet-wide, the apparent buffering is the location, not the dam

The most direct test is on streamflow. Meteorological drought (Standardized Precipitation Evapotranspiration Index of 12 months, SPEI-12; the sustained forcing is documented in Supplementary Fig. S1) propagates into streamflow drought (Standardized Streamflow Index of 12 months, SSI-12) more gently in dammed basins, the visual signature of buffering (Figure 2, panel a). But this flattening tracks where dams sit, not what they do. If the dam were doing the buffering, transmission would flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches of the same basins (differential slope -0.20 upstream versus -0.16 downstream; Figure 2, panel b). The direct within-basin test confirms the contrast is null: downstream minus upstream is $+0.04$ with a paired permutation $p = 0.72$ (Supplementary Table S25), and a variant giving pseudo-treated controls an elevation-partitioned gauge split leaves the null standard deviation unchanged, so the scheme is not compressing it ($p = 0.73$; Supplementary Table S48). The national gap of -0.18 accordingly collapses to about -0.05 once basins are compared within the same climate region, itself permutation-null ($p = 0.65$; Supplementary Table S26), and in the high-supply south, where buffering should be easiest, the upstream and downstream gaps are identical. The pooled gap does not survive randomization inference ($p_{\text{perm}} = 0.39$), and the decomposition confirms the signal dies only when each climate region carries its own baseline transmission (Supplementary Fig. S3 and Table S4).

Three independent routes through the aridity confound agree. Hard-balancing aridity makes even the *apparent* buffering vanish (intent-to-treat, the all-gauges estimand, -0.02 against -0.18; upstream -0.05 against -0.20; $p > 0.85$; Supplementary Table S17), a point-estimate shift of about two primary standard errors rather than variance inflation alone. That fit is fragile and we treat it as corroboration only: at an effective control sample of 19 its minimum detectable effect is near 77% of baseline, its weights are concentrated, and while re-balanced leave-one-out keeps the estimate near zero for seven of the eight top-weighted controls, dropping the most-weighted basin returns

-0.184 with balance restored (Supplementary Table S45). No variant turns the null into detected buffering, since every value lies inside the primary permutation null. Second, transmission is genuinely non-linear in aridity (quadratic aridity-by-forcing term $p < 10^{-5}$), and admitting that curvature collapses the apparent intent-to-treat buffering to -0.06 ($p = 0.43$; Supplementary Table S52). Third, the within-region comparison needs no aridity balance at all. The placebo and the within-region estimator therefore carry the result; the hard balance and the parametric adjustment, which interpolates within support (Supplementary Fig. S8), corroborate it.

Neither the index nor the specification manufactures the null. On log flow the gaps stay null ($p \geq 0.42$) and downstream relative variability is uncompressed (Supplementary Table S19). Re-standardizing SSI on the 1990–2020 baseline leaves every estimate essentially unchanged (contrast $+0.02$, $p = 0.89$; Supplementary Table S49), so the megadrought-heavy reference window is not distorting the slopes, and replacing the year fixed effects with a cubic national time trend, retaining the temporal drought signal they absorb, reproduces every slope (all permutation-null; Supplementary Table S47). Missing streamflow is drought-selective (34% of drought gauge-months fail the completeness rules against 14% otherwise), but the truncation is shared by treated and control gauges and nearly identical across gauge classes (33% downstream versus 35% upstream), so it thins the dry tail for every slope without manufacturing the contrast (Supplementary Table S50). Nor does the pooled slope hide early buffering: at drought onset (2010–2014, storage near-full) the upstream and downstream gaps are identical (-0.43 both), and neither phase shift nor its contrast is significant under permutation (Supplementary Tables S24, S27). A dry-tail effect is likewise absent: the quadratic SPEI interaction is null ($p = 0.65$) and drought-month refits leave every estimate and the contrast null (contrast $p = 0.49$; Supplementary Tables S28, S29).

The placebo's comparability assumptions survive testing. Upstream gauges sit 0.54 km higher, and elevation genuinely flattens transmission among control gauges (-0.21

per km, $p < 0.001$, $n = 84$, from a positive baseline of 0.53; Supplementary Table S2), but the gradient works *against* the finding: it alone would make upstream look more buffered, and the gap times that sensitivity over-explains the observed -0.04 difference, leaving at most 0.07 of downstream-specific buffering, 13% of baseline (Supplementary Table S20). Cumulative scale behaves the same way, flattening transmission among controls (+0.08 per log discharge, $p < 0.001$) while the paired basins differ little in scale. Fitting the placebo net of each gradient leaves every effect null and the contrast unchanged (elevation-adjusted $D = +0.02$; discharge-adjusted $D = -0.03$; Supplementary Tables S31, S33). Local relief, separating bedrock from alluvial reaches, likewise leaves it unchanged, moving the downstream estimate toward the upstream value rather than away (Supplementary Table S22); the snowmelt confound is excluded on the differential estimand across four specifications; and the exposure error implied by the shared subcuenca forcing is about 0.01 units, asymmetric in the direction that works against downstream-specific buffering (Supplementary Table S38).

Gauge classification survives three validations. Against the DGA hydrographic hierarchy, 67 of 70 downstream gauges lie in a sub-cuenca named for the dam's own river; excluding the three exceptions leaves the contrast null ($D = +0.04$), as does excluding the sixteen gauges within 100 m of the dam ($D = +0.12$), both moving the downstream estimate toward zero rather than toward buffering (Supplementary Table S32). The mirror-image concern is cascade contamination, an upstream gauge itself below *another* dam (Ralco above Pangue; the Laguna del Maule works above Colbún; La Laguna above Puclaro), whose flattened transmission would be real regulation misread as siting. Screening all 35 upstream gauges, carried by 16 basins (one holds upstream but no usable downstream gauge, so it enters the upstream panel but not the 15-basin paired contrast), the placebo survives every tier: excluding the three flagged within their own sub-watershed gives upstream -0.245 and $D = +0.08$; adding the documented historical cascades, ten gauges in all, gives -0.204

and $D = +0.04$; and the conservative cuenca-wide bound, stripping 20 of 35 gauges, gives -0.21 and $D = +0.05$, every contrast permutation-null (Supplementary Table S43). The direction of change is diagnostic: were the upstream flattening hidden regulation, removing flagged gauges should steepen the placebo toward zero, and instead it deepens. Routed topology closes the twin concern that elevation-classified upstream gauges sit on parallel, disconnected tributaries: snapping every gauge and dam to the HydroRIVERS network (median snap 264 m) and tracing flow paths, the strictly network-validated same-channel contrast is *positive* (down -0.24, up -0.36, $D = +0.12$, $p = 0.18$), the unregulated reaches more attenuated than the regulated ones (Supplementary Table S46). The contrast's power is disclosed rather than assumed: with 15 paired basins it detects only gaps above 0.28 SSI units, about 52% of baseline, so moderate downstream-specific buffering is not excluded (Supplementary Table S34).

A skeptical reading asks whether the flat contrast hides a real buffer masked by the extraction gradient, since downstream valleys carry far more irrigation and drought-intensified extraction would steepen the downstream slope. The gradient is real, cropland within 5 km of a downstream gauge averaging 0.20 of the buffer area against 0.08 upstream, but local extraction does not predict the transmission slope among control gauges ($p \geq 0.41$), the point estimate carries the sign opposite the masking mechanism, and the gradient times that sensitivity predicts a steepening of about 0.01 SSI units, two orders of magnitude below the minimum detectable effect (Supplementary Table S36). Re-fitting the placebo net of a SPEI-by-local-cropland interaction, with a treated-specific term so treated reaches are not held to the control-identified sensitivity, leaves the contrast null (-0.03, $p = 0.83$; Supplementary Table S37). Cropland is a static proxy, but a mechanism that leaves no imprint across a two-and-a-half-fold extraction gradient cannot hide a material buffer.

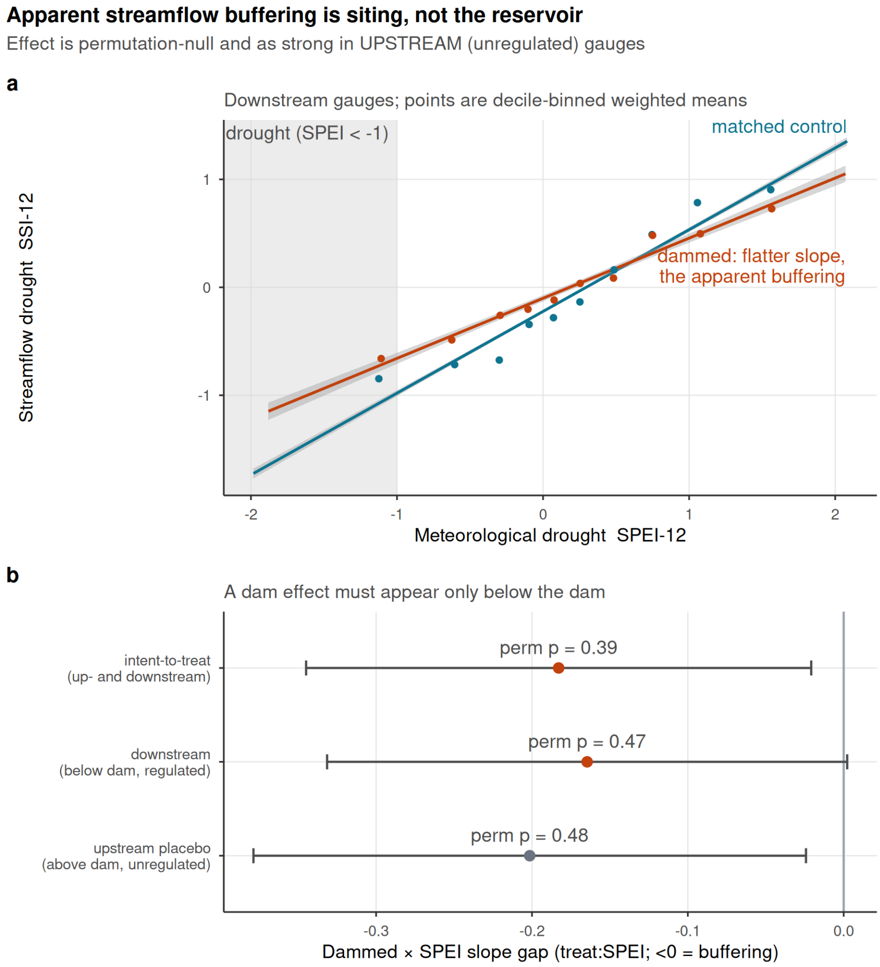

Figure 2: Fleet-wide, streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); weighted least-squares fits with 95% bands. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo, with randomization-inference p-values. All are permutation-null and the upstream placebo is as strong as downstream, so the attenuation is siting, not the dam.

3.3 The powered nulls exclude operational buffering; demand pathways are bounded

The same answer holds across the remaining pathways (Figure 3, Table 1): every estimate that carries evidential weight brackets zero, from cropland-area expansion (-0.69 ha km^{-2}, 95% CI $[-2.4, 1.1]$) and its forcing-conditioned slope gap ($p_{\text{perm}} = 0.91$). That slope gap carries a structural caveat: central Chile's cropland is dominated by perennial orchards and vineyards, which cannot flex with annual SPEI, so a null interaction is expected on rigidity grounds alone, and the informative cropland test is the dynamic event study below. The evapotranspiration pathway is excluded rather than counted as a null: whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the post-hoc orchard stratum relies on MODIS MOD16, whose documented bias toward zero over irrigated land, a product property identified independently of our results, disqualifies its null (Supplementary Table S21, Fig. S7).

The null is informative but its power is concentrated in streamflow. With about 21 dammed clusters the design cannot claim exact equivalence to zero but rules out operational buffering above 46 to 59% of baseline, and a positive control confirms it: injected buffering is recovered one-to-one and detected beyond that threshold (Supplementary Fig. S2 and Table S3). This is a wide null: a 30 to 40% reduction in transmission is fully compatible with the data, so the finding is no *large* operational buffering, not none. It also covers a record ending in March 2020, spanning the first decade of the megadrought but not the 2019–2022 peak deficits, which the storage and cropland pathways carry to 2024. Only streamflow excludes material effects at operationally relevant magnitudes: the cropland and evapotranspiration designs have minimum detectable effects far above baseline (7,330% and 237%; Supplementary Table S1), so their nulls corroborate rather than independently establish absence.

Pooling the fleet, moreover, hides a heterogeneity by storage capacity. We label this split exploratory at the outset: it was prompted in review, it is a subgroup analysis of seven treated units inside a large robustness battery, and no multiplicity adjustment is applied, so we read it as hypothesis-generating. Split by the carryover ratio on the buffering sample of 17 basins, the seven carryover reservoirs show a downstream-specific buffering the seasonal ones do not: slope -0.46 downstream versus -0.23 upstream, a contrast of -0.23 ($p = 0.014$; Supplementary Table S35), essentially unchanged on the six strictly paired basins ($D = -0.234$, $p = 0.004$; Supplementary Table S39) and surviving the elevation-partitioned null ($p = 0.025$; Supplementary Table S48); the ten seasonal reservoirs show no such contrast ($D = +0.18$, $p = 0.26$), their estimate being if anything positive, the direction drought-intensified diversion would produce though indistinguishable from zero. The signal is not a hydropower-scheduling artifact: excluding the two dams operated also for generation *strengthens* the estimate on the five irrigation-only carryover basins ($D = -0.31$, $p = 0.057$). Its fragility is instead statistical: the contrast stays negative under every leave-one-out, but the permutation p ranges from 0.004 to 0.12 and sits at or above the 0.05 convention for three of the seven exclusions (Supplementary Table S42). It is also classification-sensitive: recomputing the ratio with unregulated upstream-gauge flow keeps all seven carryover basins but moves one seasonal basin into the group, under which split the contrast is still negative but no longer significant ($D = -0.165$, $p = 0.146$; Supplementary Table S40). The direction is robust to the denominator and to single-basin exclusion; the significance is robust to neither. Within those bounds, reservoirs with enough storage to carry water across years appear to buffer multi-year drought while seasonal ones, which physically cannot, show nothing, so the aggregate null is driven by the seasonal majority.

Table 1: Reservoir-effect estimates across estimators: cross-sectional doubly-robust matched ATTs (including the water-rights induced-demand test) and forcing-interacted DiD slope gaps (treat:SPEI). The DiD row and water-rights accrual are judged by randomization-inference p-values, the design's primary inference at ~21 clusters; cluster-robust CIs over-reject here and are shown for completeness. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering.

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Cropland-area expansion	ha km^-2 added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Water-rights accrual	rights / 100 km^2 added	20.036	[4.02, 36]	0.52	null
Forcing-interacted DiD (slope-gap)	Cropland area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null

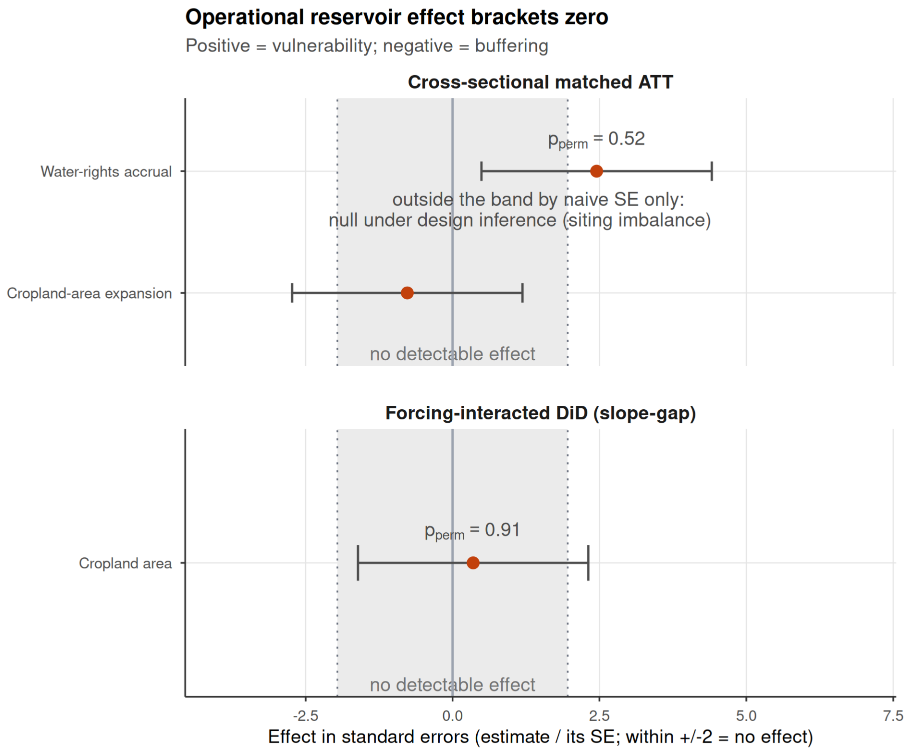

Figure 3: Dammed-versus-control effect across estimators, each expressed in its own standard errors so the axis metric is identical for every row; an estimate inside +/-2 is indistinguishable from no effect. Rows judged by randomization inference, the design's valid inference at ~21 clusters, are annotated with their p, and under that inference every estimate brackets zero. Water-rights accrual sits outside +/-1.96 by its naive standard error yet is null by the design (p_perm = 0.52), the apparent induced demand lying inside the permutation null expected from the dammed basins' unbalanced aridity (Supplementary Fig. S6).

3.4 Reservoirs mark prior vulnerability; megadrought divergence is bounded, not absent

Dammed basins carry the vulnerability reservoirs are blamed for concentrating, about 4.5 times more cropland than matched controls (10.8% versus 2.4% of basin area; Figure 4, panel a), but that gap is a *siting* signature stable across the record. A dynamic event study finds the differential trend flat before 2010 (cluster-robust $p = 0.65$, permutation $p = 0.72$) with no post-2010 divergence detected ($p = 0.81$; Figure 4, panel b), and the null is metric-invariant, likewise permutation-null on log cropland share ($p = 0.37$). Both are bounded nulls, not confirmations of parallelism: the pre-trend test detects only divergence accumulating to about half the control cropland level by 2009, and the permutation envelope widens after 2015 (Supplementary Table S14): reservoirs mark where cropland already was, and within these bounds the record is consistent with no megadrought-era expansion, though late-panel divergence remains possible.

The same holds for the most direct demand measure. In the Chilean Water Directorate (DGA) registry of consumptive water rights, dammed basins appear to accrue rights faster than controls, on both the outlier-robust count (+20 per 100 km^2; Figure 3, Table 1) and the winsorized volume (+2.5 l/s per km^2), each with an analytic interval excluding zero. This is the naive-analysis trap the design catches, since arid basins carry denser water-rights activity. The count does not survive randomization inference ($p_{\mathrm{perm}} = 0.52$) and remains null under a fully non-parametric match on the aridity-overlap subset ($p_{\mathrm{perm}} = 0.42$; Supplementary Table S7). The winsorized volume is more delicate: insensitive to the winsorization threshold on the full sample, it reaches $p = 0.005$ on the smaller aridity-overlap subset under unit-level permutation, yet those residuals are spatially autocorrelated and the effect is not significant under the governing cuenca-block permutation ($p_{\mathrm{perm}} = 0.077$), the verdict the ranked raw volumes also give. The water-rights evidence is therefore not uniformly null: the count is cleanly so, the ranked volumes are null under spatial inference, and the winsorized volume sits closer to the boundary, flagging a spatial concentration of large consumptive rights in dammed cuencas that the design can bound but cannot attribute to siting versus induced demand at this grain, and that warrants parcel-level investigation. Two institutional statements must be kept apart. Administrative closure did not mechanically silence the registry differentially: treated basins entered the drought with a higher allocated stock yet accrued new rights about 2.7 times faster than their weighted controls, and none went without new grants. But closure does make new grants a weak operationalization of total demand, because in closed basins rights change hands through private market transfers recorded outside the registry, so market-driven reallocation is invisible to the accrual measure; cropland area is the physical check against concentration at scale, and a transaction-level test would be needed to close the pathway.

3.5 Storage falls; the carryover buffer shrinks

If streamflow excludes material operational buffering and the demand pathways show no amplification within their detection limits, what matters is how much water there is to store (Figure 5). The decline is not a smooth drift: the peak steps down at the 2010 megadrought onset while the trough keeps eroding through it. Because 22 to 26 reservoir clusters are too few for cluster-robust inference, every storage p-value below is a null-imposed wild cluster bootstrap (Supplementary Table S54). Over 2005–2024 peak storage falls by 1.3 percentage points of capacity per year ($p = 0.004$) and the trough by 1.2 pp yr^{-1} ($p = 0.001$), with no detectable amplitude trend ($p = 0.98$; Supplementary Table S5); nested at the 2010 split, the peak steps down (-0.19 of capacity, $p = 0.009$) then shows no further drift, while the trough, whose step is not significant, keeps falling (-0.010 yr^{-1}, $p = 0.009$; Supplementary Table S41). Bootstrap inference tightens rather than weakens these: the peak rests at a lower equilibrium while the carryover floor still erodes. Nor is the pattern an artifact of equal-weighting a fleet whose capacities span three orders of magnitude: weighted by capacity the trough decline is steeper rather than weaker (-1.7 pp yr^{-1},

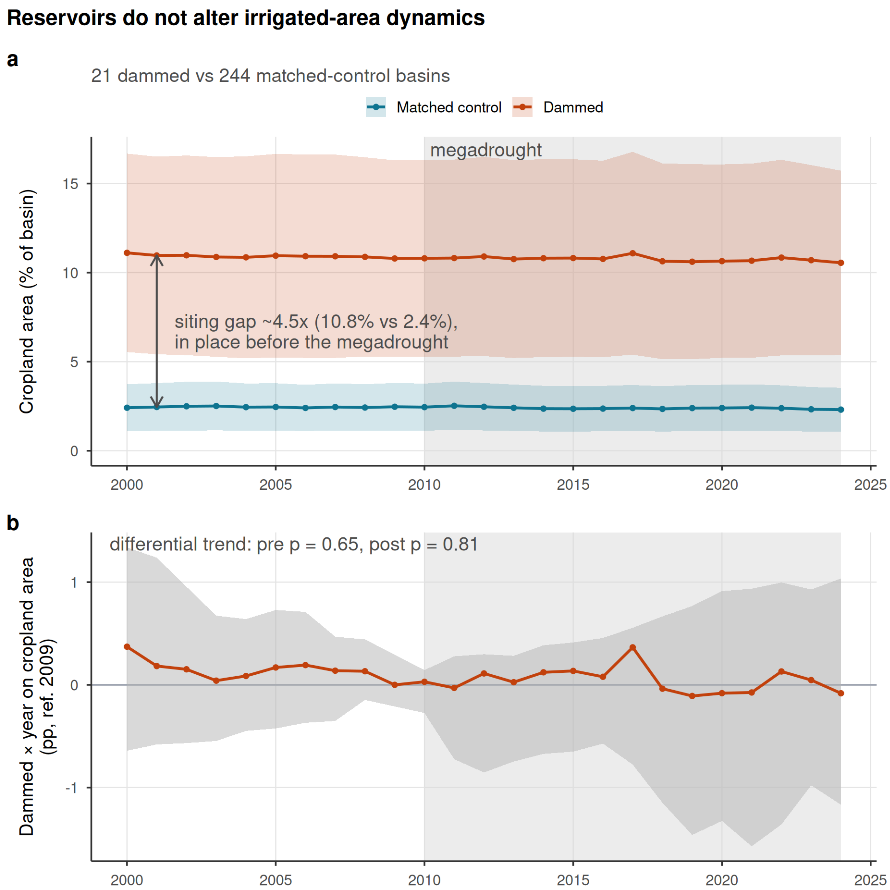

Figure 4: Cropland-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, MapBiomas class 18) with 95% bands, y-axis anchored at zero: dammed basins hold ~4.5x more cropland but the trajectories evolve in parallel. (b) Dynamic event study (dammed x year, reference 2009); the grey band is the 95% randomization-inference null envelope, cluster-robust intervals being anti-conservative at ~21 treated clusters. The envelope widens after 2015, so the flat late trajectory is a bounded null. The megadrought is shaded.

$p = 0.001$) and the amplitude widens rather than narrows ($p = 0.031$), so the flat unweighted and widening weighted amplitudes both run opposite to the narrowing that sedimentation or a failing refill cycle would produce. Weighting costs precision, though: the weighted peak trend and both weighted 2010 steps are not significant under the bootstrap, so the step evidence rests on the unweighted fit. In absolute volume the coverage-stable subset (22 reservoirs, 98% of fleet capacity) lost 88 hm^3 yr^{-1} of peak and 209 hm^3 yr^{-1} of trough storage, and restricting the trends to that subset, so mid-panel commissioning cannot tilt them, leaves the step-versus-drift separation unchanged.

The decline directly shrinks the carryover volume that buffers multi-year drought: peak storage fell from 87% of capacity before 2010 to 67% after and the trough from 40% to 25%, so the pool available to buffer a second and third dry year is materially smaller. Whether the decline is supply-driven or management-driven cannot be attributed causally, but it is directionally consistent with a supply-side reading resting on an independent decline: unregulated control streamflow fell about 2.9% per year over 2005–2020 (dammed 3.4%), larger than the 1.3 pp yr^{-1} fall in the storage stock, as expected since storage partially absorbs inflow shocks, and the megadrought runoff deficit (Alvarez-Garreton et al., 2021; R. D. Garreaud et al., 2017) is larger still in percentage terms. Sedimentation is a third candidate the percent-of-capacity metric cannot exclude by construction, since registry capacities are static nominal values and silting of active storage in these high-sediment catchments would mechanically depress the reported fraction at constant inflow. Three features bound that reading: sediment accumulation is gradual, whereas the peak decline is a level step; a large loss of live volume would compress the seasonal amplitude, which does not narrow; and a sediment-driven decline would not co-move with the independent streamflow fall, which it does. A sediment contribution to the continuing trough erosion cannot be excluded without resurveyed bathymetry, which we flag as the direct test. Storage is a mass balance, so the band alone cannot separate falling inflows from sustained releases; our direct demand outcomes point the same way, with neither cropland area nor water-rights allocation showing expansion (Figure 4, Table 1) (Li et al., 2023; Wang et al., 2024).

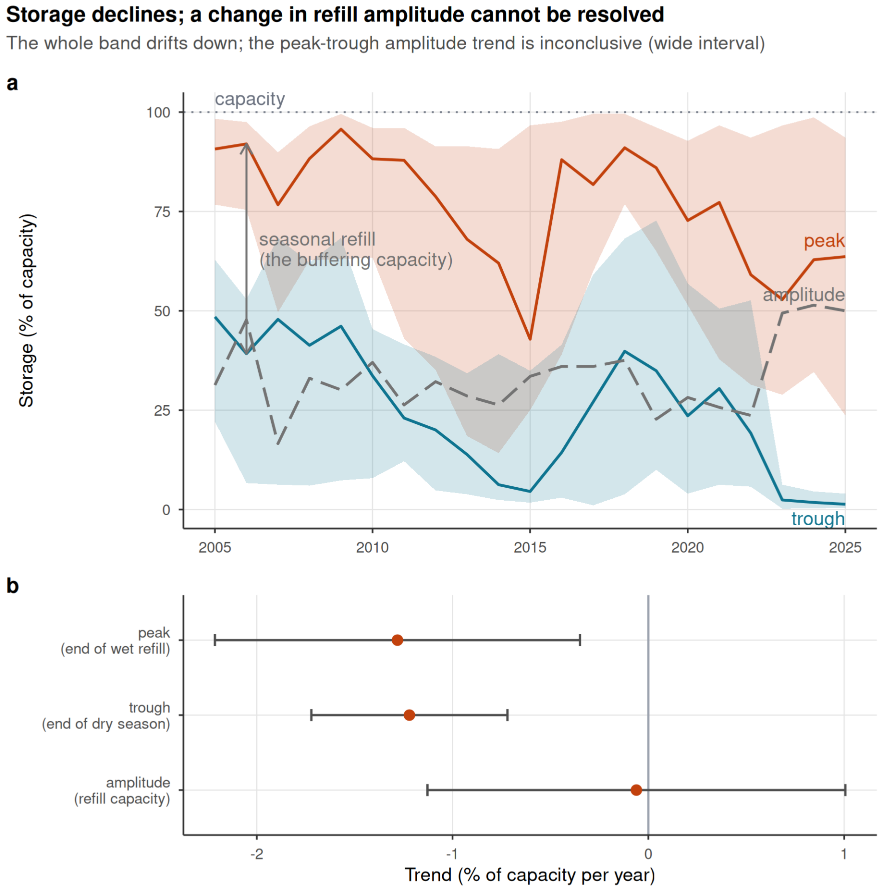

Figure 5: Reservoir storage declines; the carryover buffer shrinks with supply. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid) with interquartile ribbons, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the band drifts down while the amplitude does not narrow. (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects), with 95% cluster-robust intervals; every coefficient is confirmed by wild cluster bootstrap (Supplementary Table S54). Descriptive: pooled within-reservoir trend, no control group.

4 Discussion

Across a matched national design, multiple estimators, and a within-basin falsification test, the apparent drought buffering attributed to reservoirs is, for the seasonal majority of the fleet, a measurable siting confound. The upstream placebo is the decisive evidence: a regulation effect must appear only below the dam, and across the fleet it does not; what remains in the aggregate is the aridity gradient. The exception is itself informative: the seven carryover-capable reservoirs show a downstream-specific contrast exactly where a real regulation effect should sit, an exploratory signal bounded below, so the null is a statement about storage-limited reservoirs, not about storage as such. Dammed basins hold about 4.5 times more cropland than matched controls, but that gap is fixed across the record, with parallel pre-trends and no megadrought divergence detected within the design's power (a bounded null, not a confirmed absence). One reading fits: reservoirs mark where water-dependent vulnerability already concentrated rather than make it. This is a finding, not a failure of power: the apparent effect has an identified cause, and the positive control shows the design detects buffering when it is present.

The corollary is transferable: dammed-versus-undammed or before-after contrasts inherit the full siting confound (Lee et al., 2026; López-Moreno et al., 2009; Schilstra et al., 2024; Wu et al., 2017). The upstream placebo ports to regulated rivers with gauges above and below a dam, most directly to short, steep catchments like Chile's; in large low-gradient basins, floodplain storage and transmission losses would blur the contrast; porting it would need routing-corrected lags or naturalized-flow counterfactuals.

If the streamflow pathway excludes material operational buffering at multi-year scale and the demand-side outcomes are consistent with no amplification at their respective detection limits, what matters is the water available to store. The storage band steps down at the megadrought onset, peak and trough together, which directly shrinks the carryover volume held across years, the buffering mechanism for multi-year drought: the pool available to carry a second and third dry year is materially smaller by 2024 than it was in 2010. This is a real reduction in multi-year buffering capacity. Whether it is supply-driven or management-driven cannot be attributed causally, since the trend is descriptive and confounded with operational change; it is directionally consistent with a supply-side reading. The seasonal amplitude is unchanged within a wide interval, indicating that the reservoirs still refill to their seasonal pattern when water arrives, consistent with the carryover loss tracking falling inflow rather than a failure of the refill cycle. The powered result is the level decline, the expected mass-balance outcome of persistent inflow deficits; descriptive and confounded with management change, it supports no causal attribution but is directionally consistent with the independent decline in unregulated control streamflow, which fell faster (2.9% per year) than the storage stock (1.3 percentage points of capacity per year), as a buffer should partially absorb an inflow shock. Read against Chile's structural aridification (Alvarez-Garreton et al., 2021; R. D. Garreaud et al., 2017; Zambrano et al., 2025), accelerating Andean warming and snow loss now cutting into the snow-fed streamflow these basins depend on (Kim et al., 2025), and the global pattern of reservoirs failing to refill (Li et al., 2023; Wang et al., 2024), the binding constraint is shifting toward supply.

These findings give the contested socio-hydrological *reservoir effect* and *supply-demand cycle* (Di Baldassarre et al., 2018; Kellner, 2021) a credibly identified test, and find the protective function over-attributed to storage. Because treatment is time-invariant, we test the spatial imprint of these feedbacks, not the longitudinal loop. The 4.5-fold cropland gap could be the equilibrated legacy of demand induced at construction, so our null covers the megadrought era; historical induced demand is testable only with staggered commissioning. And the record opens in the scarcity phase, after any expansion under perceived abundance, so it bounds drought-era expansion, not the cycle as a whole. Within scope the nulls refine the theory: the cycle leaves no operational imprint during drought, so its mechanisms likely act through siting and construction rather than operation, a distinction current models omit. How far the demand-side null travels depends on governance: the siting lesson holds wherever dams are placed non-randomly; prior appropriation, which forfeits unused rights, may amplify induced demand while adaptive allocation curtails it. Whether the null is general or the mark of a fully allocated market we cannot adjudicate: rights kept accruing in dammed basins, but systems with water still unallocated could express an induced demand that ours cannot. The upstream placebo itself ports more broadly. Short, steep catchments with gauges above and below dams are common across the world's regulated mountain rivers: the California Coast Ranges and Sierra Nevada, the Spanish Sierra Nevada and Pyrenees, the Italian Apennines and Alps, the African Rift, and the Andean cordillera beyond Chile. In large low-gradient basins, the Murray-Darling, the Colorado below Lake Mead, or the Indus plains, floodplain storage and transmission losses would blur the above/below contrast and routing-corrected lags or naturalized-flow counterfactuals would be needed. The placebo method is therefore not universal, but its scope conditions are explicit and its required data, paired gauges above and below dams, exist across much of the world's irrigated mountain-front hydrology.

Chile's 1981 Water Code makes water-use rights perpetual private property, so the reflexive response has been supply-side, new dams and inter-basin transfers (Barría et al., 2021; R. Garreaud et al., 2025); the 2022 reform moves toward the adaptive allocation our results argue for (Macpherson et al., 2023) but postdates our window. Under structural aridification, building more storage is a supply-blind response to a supply problem: the buffering dams provide is a property of where they sit, not of what they do (Ho et al., 2017). Adaptation must shift toward demand management, reallocation and managed aquifer recharge (Grafton et al., 2018; Rivera-Vidal et al., 2025; Scanlon et al., 2023).

Several limitations bound these conclusions, each quantified in the Supplementary Information. The result is a bounded null: the design excludes streamflow buffering above 46 to 59% of baseline, demand-side outcomes are far less powered and the pre-trend test power-limited (Supplementary Tables S1, S14), so causal weight rests on the placebo, demand nulls corroborating; all survive the hard-balanced match, itself a fragile corroboration at its small effective sample, whose weight concentration and leave-one-out sensitivity are disclosed (Supplementary Tables S17, S45). The powered streamflow null also ends in March 2020, before the most acute 2019–2022 deficits, so the late-phase storage and cropland pathways carry that part of the record. The early-to-late phase analysis is likewise null under the design's permutation inference, with neither the downstream nor the upstream phase shift significant and their contrast far from it (Supplementary Table S27), so the flat late-phase downstream estimate is a noisy (standard error 0.36) point within its null rather than evidence of regulation; cluster-robust intervals, which the design's own rule says over-reject at about 21 clusters, overstate the phase signal. The registry measures paper rights, not wet water: Chilean basins are widely over-allocated and granted rights can sit unused or hoarded (Barría et al., 2021; Reade Malagueño & D'Odorico, 2024), so under drought, when physical availability binds, the accrual we test is an entitlement response that non-use could dampen; cropland is the physical check. The registry also records original grant dates, not market transfers: many basins have been administratively closed to new allocations for decades, so the flat accrual reflects the absence of new state grants rather than the reallocation of existing rights, which in Chile's private water market proceeds through transactions registered in local real-estate offices, not the DGA grant registry (Reade Malagueño & D'Odorico, 2024). Market-driven concentration of rights below reservoirs, if present, is therefore invisible to the accrual measure; cropland trajectories are the physical check that argues against such concentration at scale, but a transaction-level test would be needed to close this pathway. Both metrics also miss intensification within an unchanged footprint; central Chile's documented shift toward thirsty export fruit (Reade Malagueño & D'Odorico, 2024) is exactly such a transition, invisible to both; a consumption-grade test awaits thermal ET or crop-transition maps. Groundwater substitution is bounded, dammed and control water tables falling at indistinguishable rates across 213 DGA wells despite severe regional drawdown (Taucare et al., 2024); monitored wells need not sample the orchard footprints where pumping concentrates, so unmonitored drawdown could sustain a historically reservoir-induced footprint (Supplementary Table S10). In that reading the reservoir effect is not absent but displaced to the aquifer, a construction-era legacy persisting on a pool the design cannot see; the well bound and flat cropland trajectories argue against substitution at scale but do not close the pathway. The snowmelt confound, acute in these snow-fed Andean basins (Ayala et al., 2025; Götte & Brunner, 2024), is tested on the differential estimand and excluded across four specifications (Supplementary Table S12). The geomorphic check leaves the placebo unchanged, though 2.5 km relief cannot resolve aquifer storage itself, a residual caveat (Supplementary Table S22). Cascade contamination of the upstream placebo, upstream gauges that are themselves regulated by another dam, is screened against the 1,370-dam national inventory and excluded at three tiers, and removing the flagged gauges deepens rather than removes the upstream attenuation (Supplementary Table S43); the classification is likewise validated against routed network topology, where the strictly same-channel contrast turns positive, unregulated reaches more attenuated than regulated ones (Supplementary Table S46). Streamflow missingness is drought-selective but common across gauge classes, so it truncates the dry tail without generating the null contrast (Supplementary Table S50). The placebo also spans land-cover and abstraction contrasts: within paired basins, downstream valleys carry about two and a half times the cropland of upstream headwaters (per-basin means, 12 of 15 basins downstream-richer; Supplementary Table S36), and upstream reaches carry unrecorded diversions. Gauge fixed effects absorb these level differences, and the gradient cannot mask a buffer: local cropland intensity does not predict the transmission slope among control gauges (p 0.41), the point estimate carries the sign opposite to the masking mechanism, and re-fitting the placebo net of a SPEI-by-local-cropland interaction leaves the contrast null (-0.03, $p = 0.83$; Supplementary Tables S36, S37). The cropland sensitivity is identified among control gauges, so a release-sustained extraction response could exceed it; the adjusted model therefore lets treated reaches carry their own SPEI-by-cropland term, and the implied masking magnitude sits two orders below the contrast MDE either way. Administrative closure cannot force the demand null: treated basins entered the drought with higher allocated stock yet accrued rights 2.7 times faster, and parallel control cropland contradicts collapse prevention (Supplementary Table S15); what closure does do is make grant accrual a weak proxy for total demand, the registry limitation discussed above. No matched pair shares a watershed; outbound transfers would shrink the gap and imports would mimic downstream-specific buffering, which is absent (Supplementary Table S15); irrigation-only restriction leaves the result unchanged (Supplementary Table S11). MOD16's bias toward the null excludes the ET pathway from evidence (Supplementary Table S21); the demand conclusion rests on cropland and the registry, with thermal ET as future work. The 12-month accumulation targets multi-year vulnerability, not the sub-seasonal buffering reservoirs also provide, which we neither test nor dispute. The fleet is partly seasonal by design: the median treated reservoir stores 0.3 years of downstream flow, and only seven of seventeen hold at least half a year (Supplementary Table S23). This heterogeneity is informative: the seven carryover-capable reservoirs show a downstream-specific buffering the seasonal ones do not (contrast -0.23, $p = 0.014$; Supplementary Table S35), so the aggregate null is a statement about the seasonal majority, not about reservoirs that can actually carry water across years. Reservoirs with enough storage to bridge a dry year appear to buffer multi-year drought; seasonal ones, which physically cannot, do not. The signal survives the confounds we can test but not every inferential stress. It is not hydropower scheduling: excluding the two dams operated also for generation (Colbún, Lago Laja) strengthens the contrast on the five irrigation-only carryover basins ($D = -0.31$, $p = 0.057$; Supplementary Table S42). It is directionally stable under every leave-one-out, but its permutation p spans 0.004 to 0.12 across single-basin exclusions (Supplementary Table S42), and the threshold is classification-bounded: recomputing the carryover ratio with unregulated upstream-gauge flow, the available inflow proxy, keeps every carryover basin classified as carryover but reclassifies one seasonal basin into the group, under which split the carryover contrast is still negative but no longer significant ($D = -0.165$, $p = 0.146$; Supplementary Table S40). Together with the absence of any multiplicity adjustment across the paper's robustness battery, this is why the split is labeled exploratory. It bounds the generalization: the headline null applies to the storage-limited fleet, and the capacity-threshold signal is a directionally robust candidate for a rule-curve or commissioning-based test with more carryover reservoirs. Two extensions can test what our window cannot: a staggered event study of cropland around historical commissioning would probe construction-phase induced demand; a rule-curve quasi-experiment on release records, once public, would isolate operation.

5 Conclusions

At the multi-year timescale of structural aridification, then, the design detects no operational buffering above 46 to 59% of baseline across the fleet and no megadrought-era demand expansion beyond the siting legacy: for the seasonal majority, the apparent buffering is a statistical shadow of where dams sit, not of what they do, while the water available to store declines. The qualification is physical: the seven carryover-capable reservoirs, holding at least half a year of downstream flow, do show downstream-specific buffering, an exploratory but directionally robust signal, so the aggregate null describes the seasonal majority. The nulls are bounds, not zeros; effects inside them can be volumetrically material (Supplementary Table S16). But under structural aridification, with no operational buffering within the design's bounds for the storage-limited fleet, building more storage is a supply-blind response to a supply problem. Chile continues to plan new reservoir construction to expand storage, while its agricultural water demand continues to grow under perpetual private water rights that the 2022 Water Code reform only begins to address (Macpherson et al., 2023).

The alternative is not theoretical: demand reallocation through active water markets, irrigation efficiency, managed aquifer recharge, and crop transitions toward lower water footprints are available now and do not depend on rainfall (Grafton et al., 2018; Rivera-Vidal et al., 2025; Scanlon et al., 2023). Resilience to a falling supply ceiling will come from confronting demand and allocation, not from more impoundments.

6 Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.21052149). All analyses run in R, using `WeightIt`/`cobalt` for balancing and diagnostics, `fixest` for fixed-effects regression, and `sf`/`terra`/`exactextractr` for the spatial extractions; the pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (SPEI-12 forcing and baseline aridity), MODIS MOD16 (evapotranspiration), MapBiomas Chile (land cover), the CR2 network (streamflow), ERA5-Land (snow water equivalent), SRTM (elevation), and the DGA reservoir, watershed, water-rights, and well-hydrograph registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

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